

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Mounting plate, Part Number ST-749
- Ball joint vise, Part Number ST-302
- Air compressor seat removal tool, Part Number 3377416
- Valve spring tester, Part Number 5299714
- Air compressor seat installation tool, Part Number 3377415

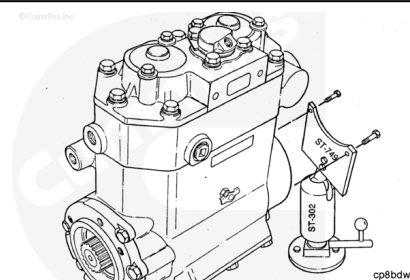
Additional Service Items

- Press
- O-ring pick

Disassemble

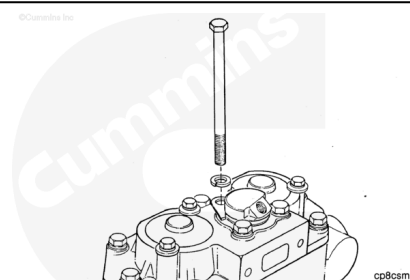
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

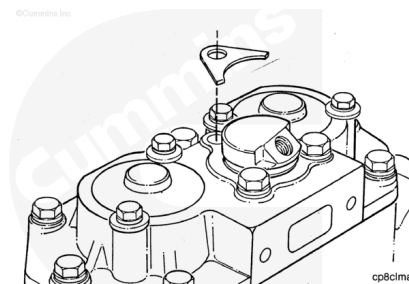


Use two 5/16 - 18 x 1 1/4 inch capscrews to mount the air compressor on the mounting plate, Part Number ST-749, which is used with the ball joint vise, Part Number ST-302.

Remove the center unloader body hexagon head capscrew and lock washer.



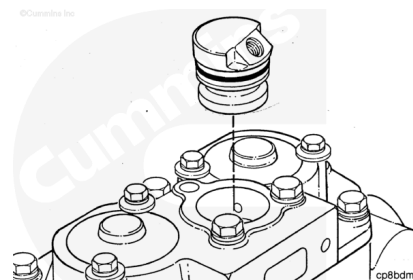
Remove the retaining clamp.



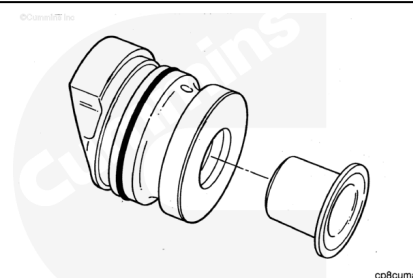
⚠ CAUTION ⚠

The unloader valve body is installed with spring tension. Carefully remove to avoid personal injury. Wear face and eye protection.

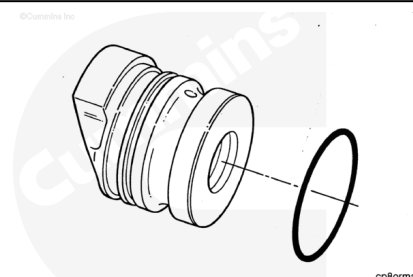
Remove the unloading valve body.



Remove the unloader valve cap, if installed.

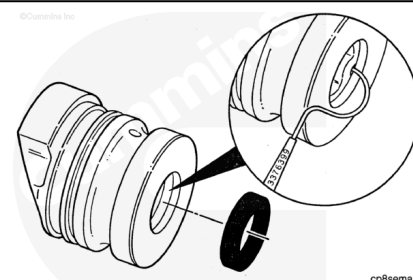


Remove and discard the o-ring seal.

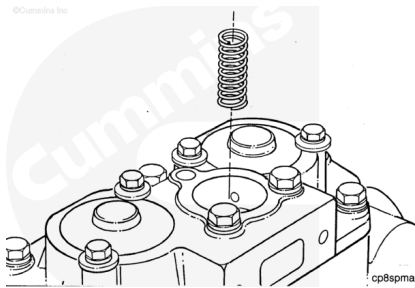


Use o-ring pick to remove the rectangular ring seal, if installed.

Discard the seal.

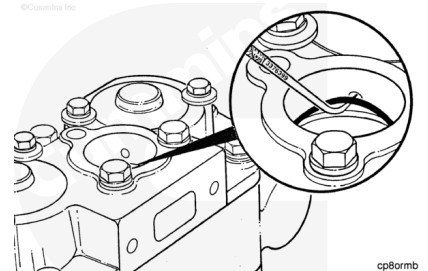


Remove the unloader valve cap spring.

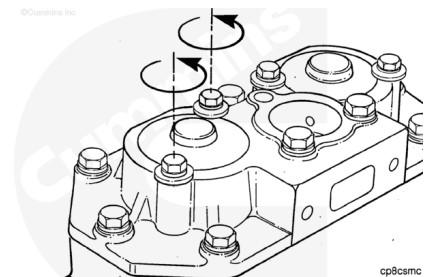


Use o-ring pick to remove the o-ring seal.

Discard the seal.



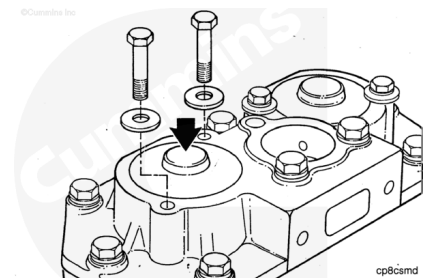
Loosen the two unloader body hexagon head capscrews.



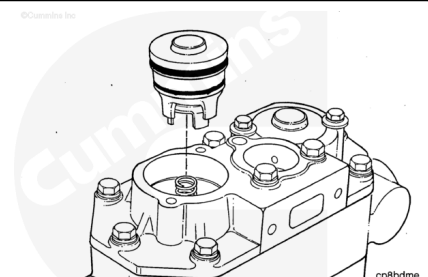
⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.

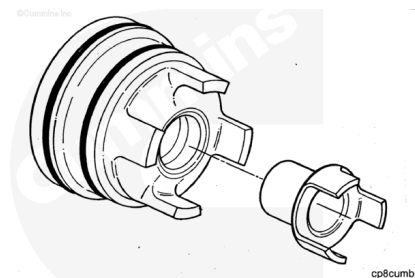
Hold down on the unloader body and slowly remove the two unloader body capscrews and the two plain washers so as to prevent the unloader body from being thrown free and causing personal injury.



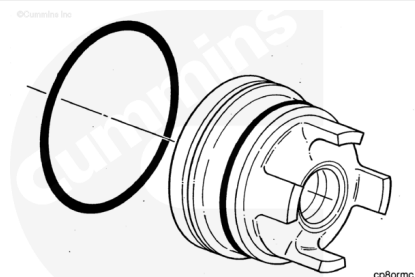
Remove the unloader valve body.



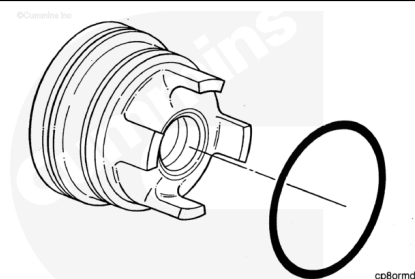
Remove the unloader cap.



Remove and discard the top o-ring seal.

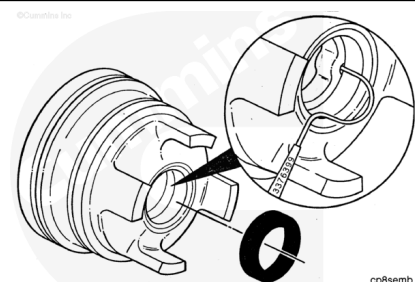


Remove and discard the bottom o-ring seal.



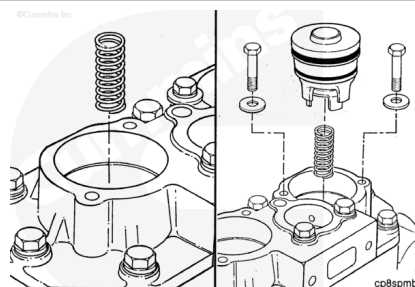
Use o-ring pick to remove the rectangular ring seal.

Discard the seal.



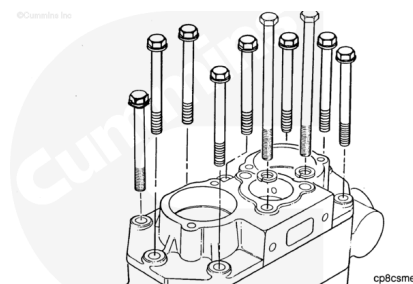
Remove the unloader cap spring.

Repeat the last eight steps to remove the other unloader body assembly.

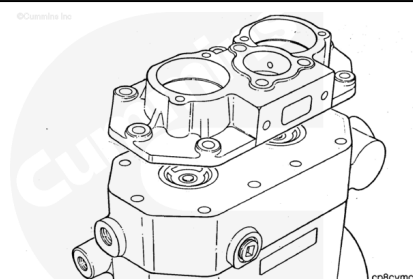


For the ST676, remove the ten remaining hexagon head capscrews and two lock washers.

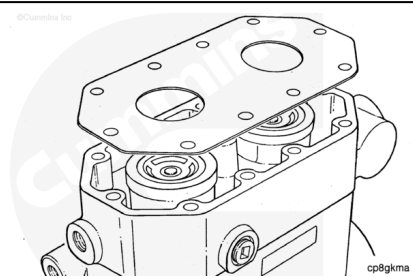
For the ST773, remove the eight captive washer capscrews and the two hexagon head capscrews and lock washers.



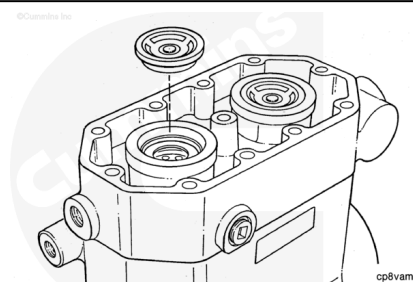
Remove the cover.



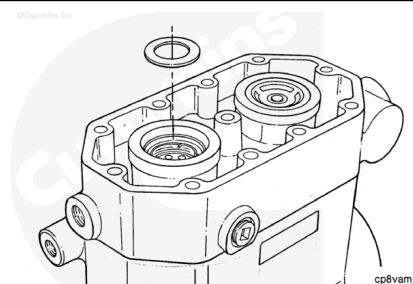
Remove and discard the cover gasket.



Remove the intake valve seat.

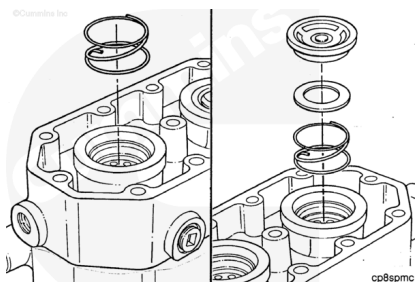


Remove the intake valve.

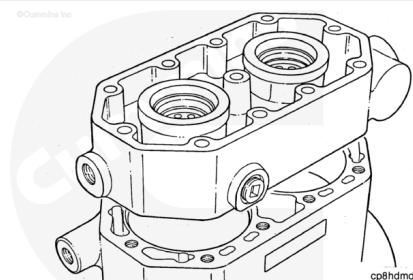


Remove the intake valve spring.

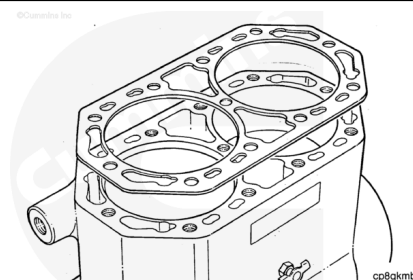
Repeat the last three steps to remove the other intake valve assembly.



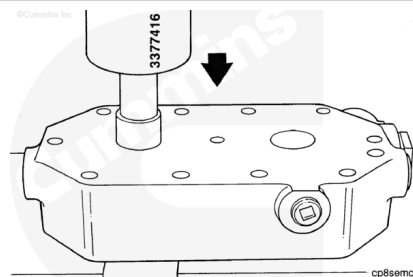
Remove the head.



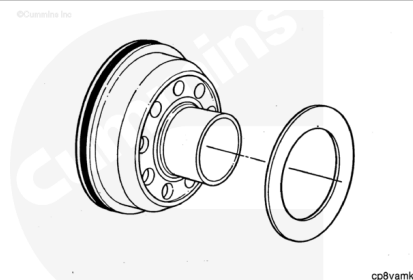
Remove and discard the head gasket.



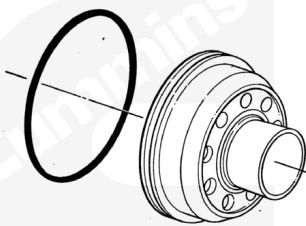
Use a press and air compressor seat removal tool, Part Number 3377416, to remove the exhaust valve seat assembly from the head.



Remove the exhaust valve.



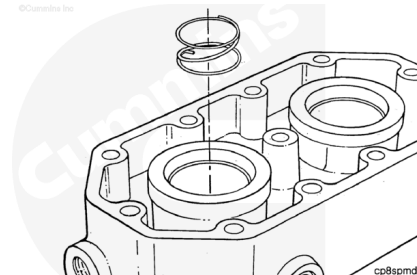
Remove and discard the o-ring seal.



cp8b0me

Remove the exhaust valve spring.

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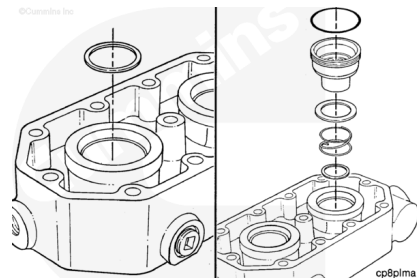


cp8spmd

Remove the wear plate.

Repeat the last five steps to disassemble the other exhaust valve seat assembly.

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cp8plma

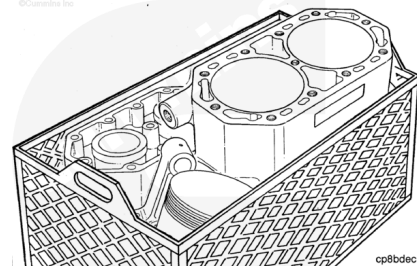
Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Soak the parts in a kerosene emulsion based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier **must** be contacted about solution concentration, temperature, and soak time.

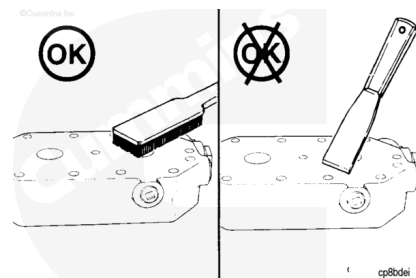
©Cummins Inc.



cp8bdec

Note : Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.

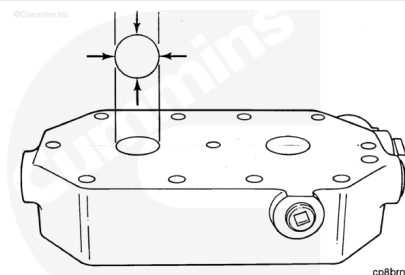
The parts can be scrubbed with a stiff nonmetallic bristle brush.



Inspect for Reuse

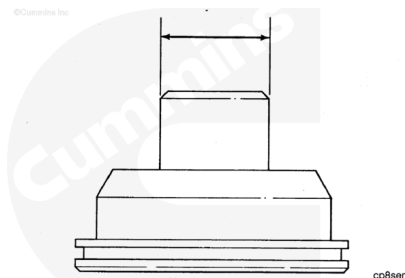
Measure the exhaust valve seat press fit diameter in the cylinder head.

Replace if worn larger than 20.688 mm [0.8145 in].



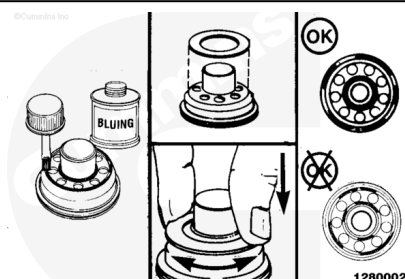
Measure the exhaust valve seat press fit diameter.

Replace if worn smaller than 20.714 mm [0.8155 in].



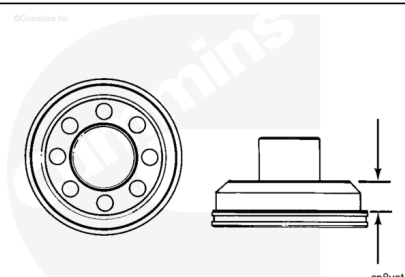
Apply bluing to the exhaust seating surface to check the seat.

If the seating surface is **not** 100 percent true, discard and replace the valve seat.



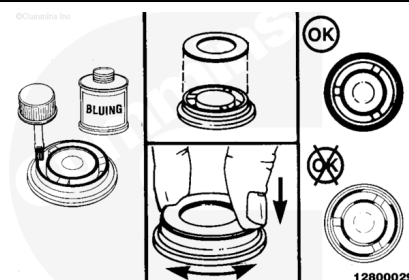
Measure the exhaust valve seat height.

If the height is less than 12.32 mm [0.485 in], replace the exhaust valve seat.



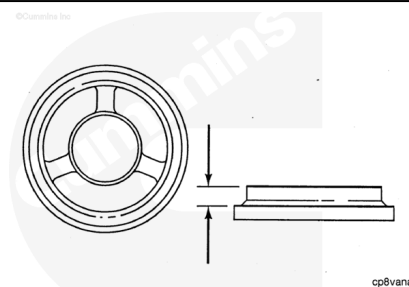
Apply bluing to the intake valve seating surface.

If the seating surface is **not** 100 percent true replace the intake valve seat.



Measure the intake valve seat height.

If the height is less than 6.86 mm [0.270 in], replace the intake valve seat.

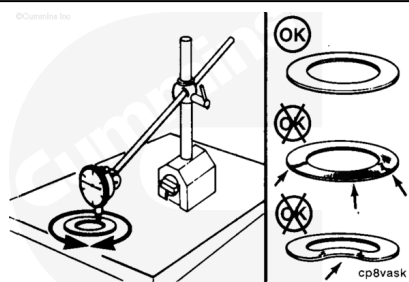


Inspect the exhaust and intake valves for cracks and damage.

Measure valves for flatness. The valves **must** be flat within 0.03 mm [0.001 in].

Replace the valves if cracked, damaged, or **not** flat.

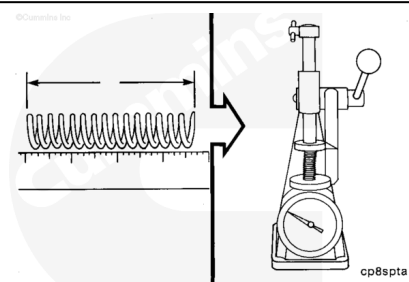
Note : Holset Engineering Co., Inc. recommends new valves be installed during rebuild.



Use valve spring tester, Part Number 5299714, to check the springs.

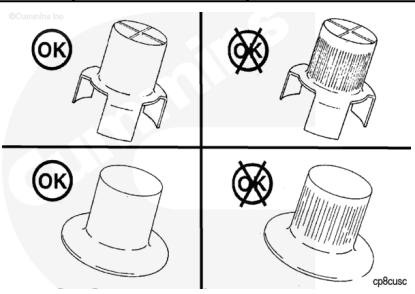
Replace if within limits listed in the Compressor Spring Force Specifications Table shown below.

Note : Holset Engineering Co., Inc. recommends new springs be installed during rebuild.



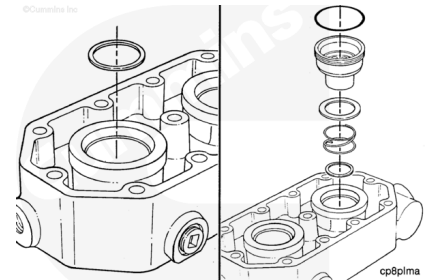
Load Required to Compress Spring to Measurement Length

Spring Part Number	Spring Use	Approx. Free Length: mm [in]	Number of Coils	Wire Diameter : mm [in]	Measure ment Length: mm [in]	Minimu m: kg [lb]	Maximu m: kg [lb]
128080	Exhaust valve	17.02 [0.670]	3.0	2.03 [0.080]	7.11 [0.280]	3.6 [8.55]	4.7 [10.35]

Load Required to Compress Spring to Measurement Length							
Spring Part Number	Spring Use	Approx. Free Length: mm [in]	Number of Coils	Wire Diameter : mm [in]	Measure ment Length: mm [in]	Minimu m: kg [lb]	Maximu m: kg [lb]
190334	Intake valve	12.70 [0.500]	2.8	1.58 [0.062]	7.11 [0.280]	0.35 [0.65]	0.5 [1.10]
150631	Unloadin g valve (naturally aspirate d) center unloadin g valve - twin	41.91 [1.650]	11.5	2.03 [0.080]	24.89 [0.980]	14.5 [32.00]	17.2 [38.00]
3023101	Unloadin g valve (all turbocha rged engines)	41.91 [1.650]	10.8	1.65 [0.065]	24.89 [0.980]	5.9 [13.00]	7.7 [17.00]
3049553	E-Type unloader valve	41.91 [1.650]	11.25	1.93 [0.076]	24.89 [0.980]	10.4 [23.00]	12.2 [27.00]
800399-XW	Unloader	17.02 [0.67]	6	1.04 [0.041]	6.60 [0.260]	N/A	2.54 [5.60]
					10.03 [3.95]	1.71 [3.78]	N/A
802000-FZ	Intake and exhaust	10.16 [0.40]	4.25	0.79 [0.031]	4.57 [0.18]	N/A	0.36 [0.80]
					5.99 [0.275]	0.20 [0.45]	N/A
3054489	Exhaust valve	21.49 [0.846]	4.5	2.54 [0.100]	15.21 [0.599]	3.88 [8.55]	4.74 [10.45]
Inspect the upper part of the unloader cap where the rectangular ring seal seats for scoring. Replace if scored.							

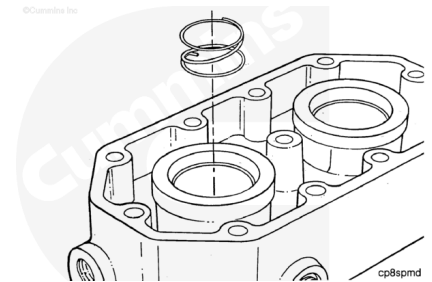
Assemble

Install the wear plate.

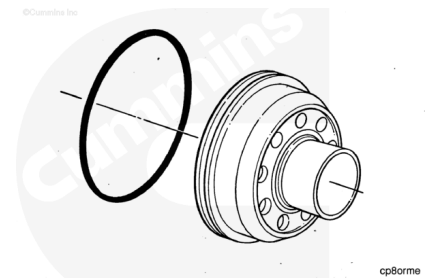


Install the exhaust valve spring with the tang end **down**.

Note : Holset Engineering Co., Inc. recommends new springs be installed during rebuild.

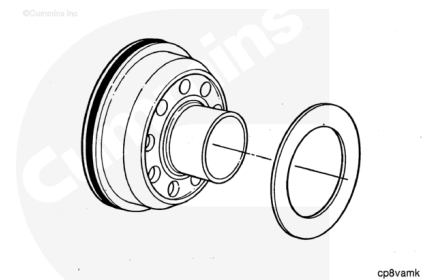


Install a new o-ring seal.

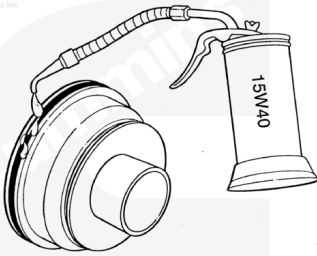


Install the exhaust valve.

Note : Holset Engineering Co., Inc. recommends new exhaust valves be installed during rebuild.



Use clean 15W-40 oil to lubricate the seal.

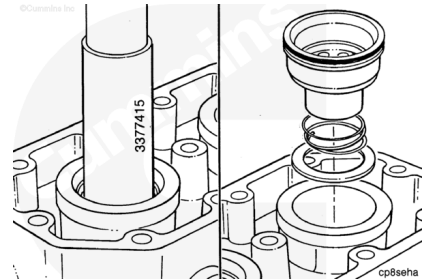


cp8orwa

Note : Do not use excessive pressure on the exhaust valve seats, to do so can distort the valve.

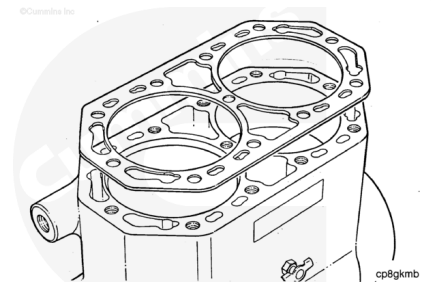
Use a hand press and an air compressor seat installation tool, Part Number 3377415, to press the exhaust valve seat into the head.

Repeat the last six steps to install the other exhaust valve assembly.



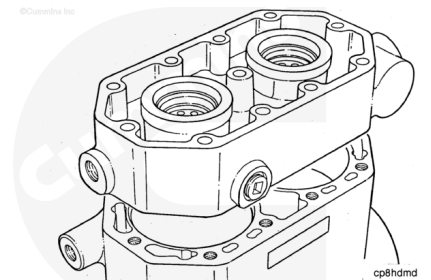
cp8seha

Install a new cylinder head gasket.



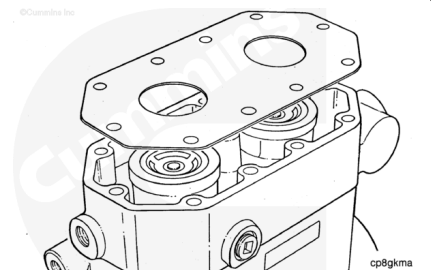
cp8gkmb

Install the head.



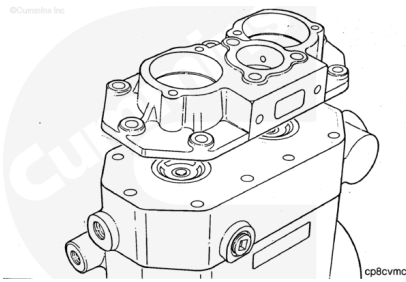
cp8hdmd

Install a new cover gasket.



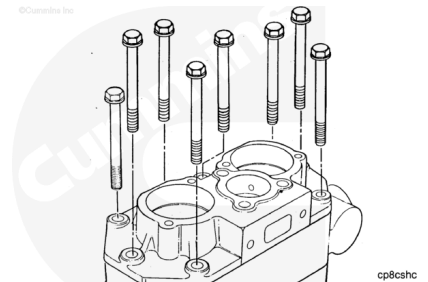
cp8gkma

Install the cover.

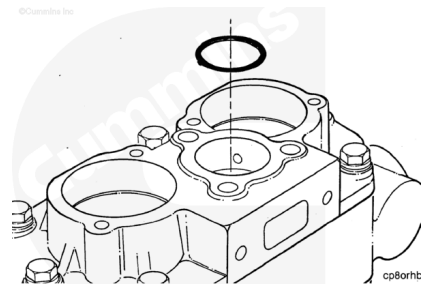


For the ST676, install the eight hexagon head capscrews.
Finger tighten.

For the ST773, install the eight captive washer capscrews.

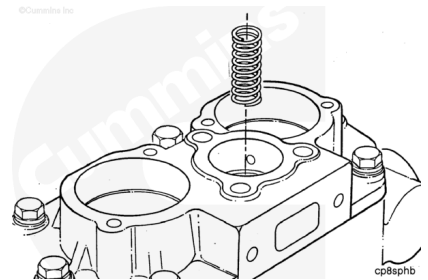


Use o-ring pick to correctly install the new o-ring seal.



Install the unloading spring.

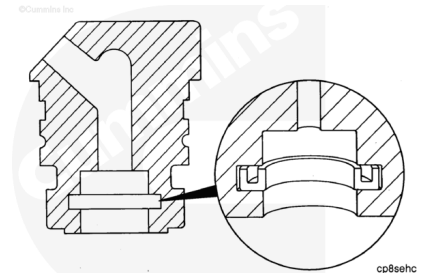
Note : Holset Engineering Co., Inc. recommends new springs be installed during rebuild.



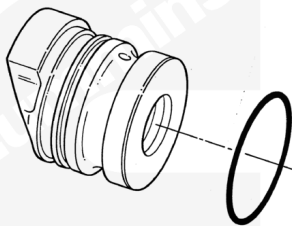
⚠ CAUTION ⚠

The rectangular ring seal must be installed with the grooved side up; failure to do so will result in air system damage and brake malfunction.

Use o-ring pick, Part Number 3376399, to correctly install the new rectangular ring seal, if the unloader body has a cavity for this o-ring.

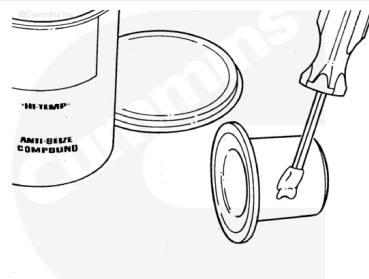


Install a new o-ring seal.



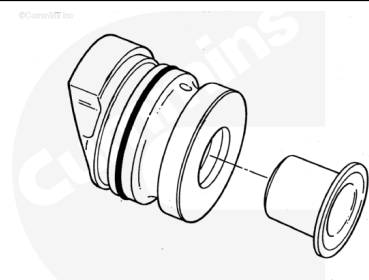
cp8orma

Use anti-seize compound to lubricate the outside diameter of the unloader valve cap.



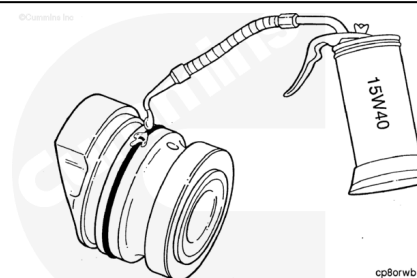
cp8cuwa

Install the cap into the unloading valve body.



cp8cuma

Use clean 15W-40 oil to lubricate the seal.

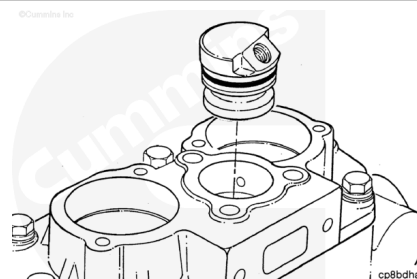


cp8orwb

⚠ WARNING ⚠

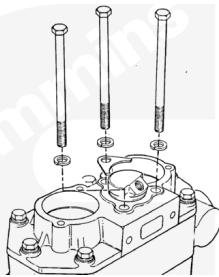
Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.

Install the body into the cover.



cp8bdha

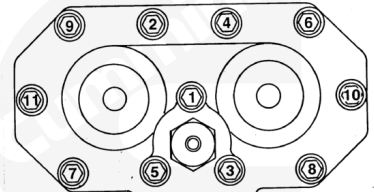
Install the retaining clamp and the three lock washers and the three hexagon head capscrews.



cp8cshd

Tighten the capscrews in the sequence illustrated.

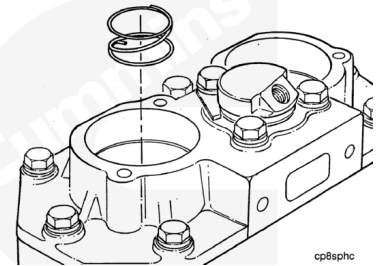
Torque Value: 41 n•m [30 ft-lb]



cp8csog

Install the intake valve spring with the tang end **down**.

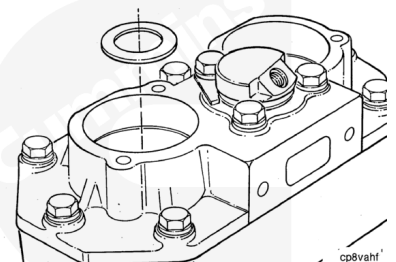
Note : Holset Engineering Co., Inc. recommends new springs be installed during rebuild.



cp8sphc

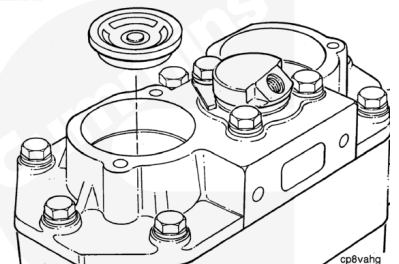
Install the intake valve.

Note : Holset Engineering Co., Inc. recommends new intake valves be installed during rebuild.



cp8vahf

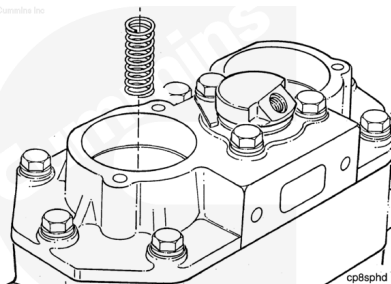
Install the intake valve seat with the flange end **up**.



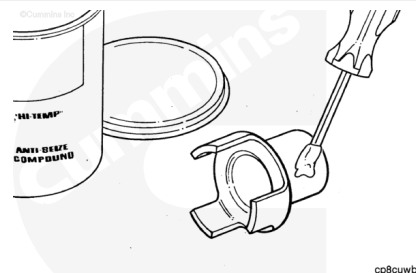
cp8vahg

Install the unloader cap spring.

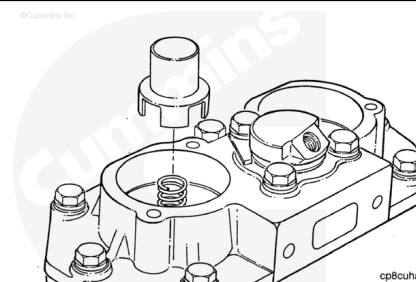
Note : Holset Engineering Co., Inc. recommends new springs be installed during rebuild.



Use anti-seize compound to lubricate the outside diameter of the unloader valve cap.



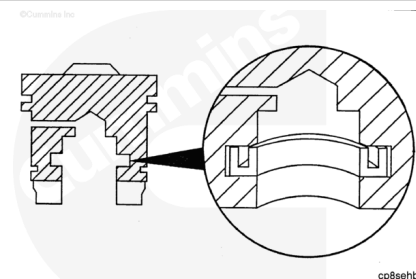
Install the cap into the cover and make sure the three tangs are in the three slots of the intake valve seat.



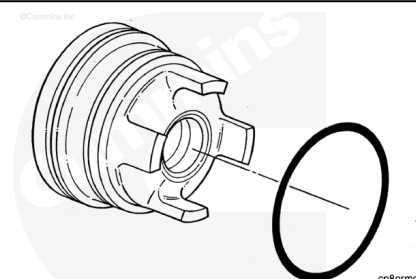
⚠ CAUTION ⚠

The rectangular ring seal must be installed with the grooved side up; failure to do so will result in air system damage and brake malfunction.

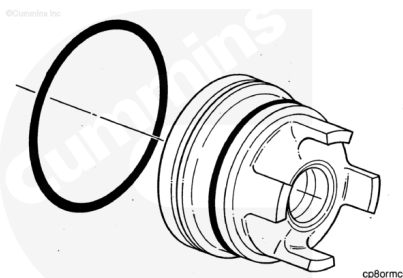
Use o-ring pick to correctly install the new rectangular ring seal.



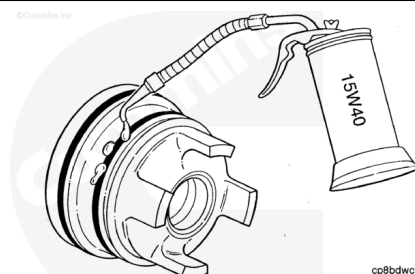
Install a new bottom o-ring seal.



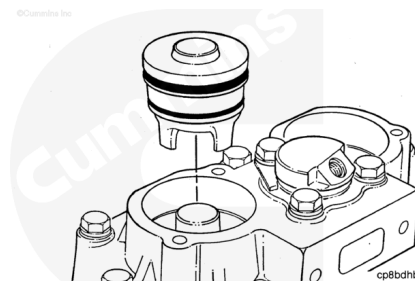
Install a new top o-ring seal.



Use clean 15W-40 oil to lubricate the seals.



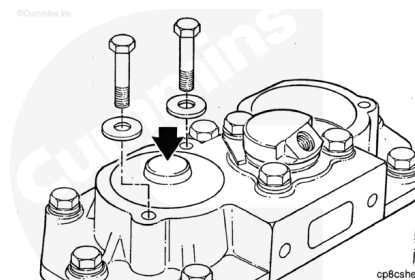
Install the unloading body into the cover.



⚠ CAUTION ⚠

The unloader body is installed with spring tension. It must be installed as described to prevent personal injury.

Hold down on the unloading body firmly. Install the capscrews and plain washers so as to prevent the unloader body from being thrown free and causing personal injury.



Tighten the capscrews.

Torque Value: 14 n•m [124 in-lb]

Repeat the last thirteen steps to install the other unloading body assembly.

